Yongchun Li - Week 10 Report (*1378156*)

1. Tasks Accomplished This Week

1. Attended daily stand-up meetings, delivered personal contribution presentation(1 h).
2. Participated in the weekly team retro & next‐week planning (2 h)
3. Preliminary Research on the Email Notification Module (5h)
   (1) Using the Resend Email API in conjunction with a domain-bound mailbox managed via Cloudflare.
   This approach demonstrated the highest stability and required no local SMTP configuration. The API-based interaction is clean and easy to integrate. However, it comes with a limitation of 100 free emails per day, and its tight coupling with the domain registrar potentially compromises future flexibility and portability of the system.
   (2) Leveraging the native JavaMail API to send emails via the SMTP protocol.
   This solution involves minimal external dependencies and offers a straightforward implementation path. Nonetheless, it demands significant manual configuration efforts, such as setting up SSL/TLS, authentication, and connection management. Moreover, email sending and receiving are handled by entirely different protocol stacks (SMTP for sending, IMAP/POP3 for receiving). If future requirements involve inbound email processing, additional components and configurations will be necessary, leading to increased maintenance overhead.
   (3) Utilizing Spring Boot's JavaMailSender, configured with a third-party SMTP provider.
   This option builds on top of the JavaMail API, providing a higher level of abstraction and better integration within the Spring ecosystem. It supports advanced features such as asynchronous sending and templated email generation, resulting in improved developer productivity and maintainability. However, since it still relies on external SMTP services, the solution must address potential constraints such as rate limiting, sender reputation, and spam detection.

Ultimately, I opted to implement the solution using Spring Boot's JavaMailSender, as it offers a balanced trade-off between ease of integration, maintainability, and scalability. Additional features such as templated emails and message queuing will be considered in future iterations depending on system requirements.

2. Key Issues Encountered

After discussion, it was observed that the backend team was relatively large, which made it challenging to fully allocate meaningful tasks during the planning and prototyping phases.

3. Plan for Week 11

- ✧ Continue developing of Epic 1, 2, 9
- ✧ Make slides for final pre as well as support those presenting members